



NITI Aayog

DRAFT

SECTORAL GLANCE

Electronics

Manufacturing in India

DRAFT

Sector at a Glance

Electronics

Manufacturing in India

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview



The Indian electronics industry demonstrates robust growth, expanding its global share to 3.75%.

Contributing 3-4% to India's GDP, it employs 1 million people.

Facing challenges like import dependency, solutions involve R&D, infrastructure enhancement, and emphasizing export competitiveness for global success.

The domestic electronics industry has posted strong growth over the past decade (up to fiscal 2022), with its global electronics manufacturing share expanding to 3.75% in fiscal 2022 from 1.30% in fiscal 2012¹, as per industry estimates. In fact, the sector contributes 3-4% share of India's gross domestic product in fiscal 2022², as per the India Brand Equity Foundation. Also, the industry has played a vital role in job creation, employing nearly 1 million people and accounting for ~6% of the country's manufacturing sector workforce.

Demand for electronics has accelerated in the past five years, on the back of rapidly evolving technology and deepening internet penetration, which has sharply increased demand for electronic devices. Smartphones, laptops, tablets, etc. have become essential devices for the urban as well as rural populace. According to the Ministry of Electronics and Information Technology, demand for electronics consumption has grown at ~11% CAGR for the past 5 years to ~\$140 billion in fiscal 2022³.

Also, as per the quick estimates, Press Information Bureau estimates for selected major commodities in March 2023⁴, electronic exports have now become the sixth-largest export commodity group.

The Indian electronics industry grapples with challenges such as high import dependency for components, inadequate infrastructure, and cost disadvantages. To address these issues, a strategic shift is imperative. Strengthening the domestic component ecosystem, investing in research and development, improving ease of doing business, emphasizing export competitiveness, and establishing electronics specific SEZs are crucial steps toward fostering a self-reliant and globally competitive electronics manufacturing sector in India.

¹ [Press Information Bureau, 2023](#)

² [India Brand Equity Foundation, 2023, Electronic and Computer Software Industry and Exports](#)

³ [Ministry of Electronics and Information Technology, 2023, Annual Report](#)

⁴ [Press Information Bureau, 2023](#)

Snapshot of electronics sector

In FY22, India demonstrated a robust electronics industry, with rising domestic production and exports, signaling a reduced dependence on imports.

			
Consumption	Domestic production	Exports	Imports
Rs 10,407 billion in FY22	Rs 6,408 billion in FY22	Rs 1,098 billion in FY22	Rs 5,097 billion in FY22
Past 5 years: CAGR 11%	Past 5 years: CAGR 15%	Past 5 years: CAGR 22%	Past 5 years: CAGR ~2%

Source: Ministry of Electronics and Information Technology⁵

Note: Domestic production includes assembly of imported product

The Indian electronics industry has witnessed a steady decline in overall import of electronic goods over the years. Imports account for approximately 10.5% of the country's total imports in fiscal 2023 vis-à-vis ~14.0% in fiscal 2021⁶. Among the various categories within the electronics industry; electronic components, computer hardware, and industrial electronics have emerged as the segments with highest import value

Figure 1: Imports of Electronics grew at a 5-year CAGR of 12%

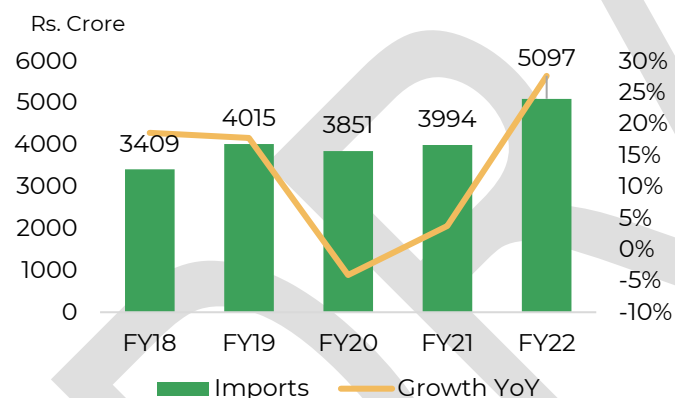
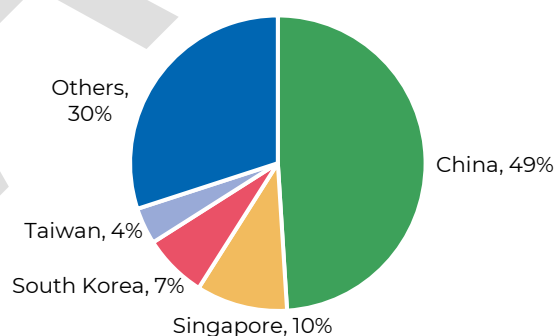


Figure 2: Key Countries for Electronics Imports



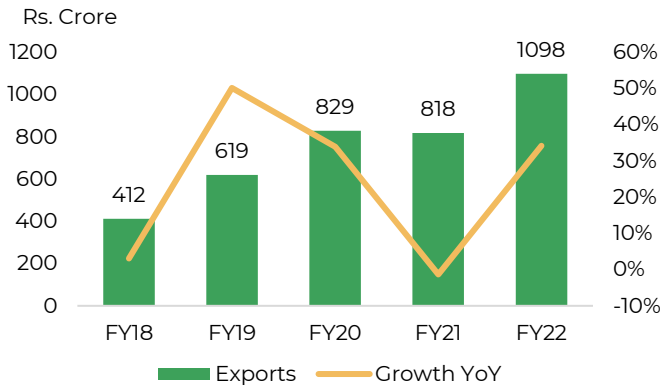
Source: Ministry of Electronics and Information Technology in India⁷ Note: As of FY23

⁵ Ministry of Electronics and Information Technology, 2023, Annual Report

⁶ India Budget, 2023, Economic Survey 2022-23 Statistical Appendix

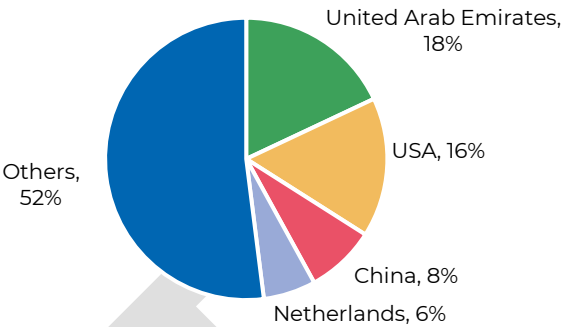
⁷ Ministry of Electronics and Information Technology, 2023, Annual Report

Figure 3: Exports of Electronics grew at a 5-year CAGR of 22%



Source: Ministry of Electronics and Information Technology in India8 Note: As of FY23

Figure 4: Key Countries for Electronics Exports



Sub-sectoral trends⁴



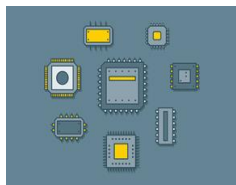
Mobile Phones

India's mobile phone segment has witnessed remarkable growth, positioning the country as the second largest mobile handset manufacturer globally. According to Ministry of Electronics and Information Technology, India has become the #2 smartphone market in the world⁵, solidifying its status as the fastest growing smartphone market globally.



Industrial electronics

The domestic industrial electronics industry encompasses various segments, including power electronics, DC/AC converters, material handling and industrial robots. Key application segments of this the industry are process control equipment, testing and measuring equipment, power electronics equipment, automation, and analytical instruments.



Consumer electronics

The consumer electronics industry in India comprises a wide range of electronic devices designed for everyday use, including televisions, washing machines, air conditioners, refrigerators and more. This growth is propelled by factors such as new product launches, technological advancements, and competitive pricing strategies, making electronic devices more accessible to a broader consumer base. Also, 100% foreign direct investment is permitted in consumer electronics manufacturing sector via the direct route, and manufacturers can avail capex subsidy of 20% for investments in Special Economic Zones (SEZs) and 25% in non-SEZs under the government's Modified Special Incentive Package Scheme.

Sub-sectoral trends⁹



Electronic components

Electronic components are essential building blocks of electronic devices and systems, encompassing resistors, capacitors, diodes, transistors, integrated circuits, and more. The electronic components industry in India is poised for significant growth, in line with the global trend. In India, the demand for electronic components is primarily driven by mobile phone, consumer electronics and industrial electronics segment, which together account for 82% market share for electronic components in India.



IT/Computer Hardware

The Indian IT hardware industry is growing rapidly, driven by advancements in artificial intelligence, machine learning and internet of things, and emergence of large-scale data centers. With its technical expertise, skilled workforce, and government schemes like The Production-Linked Incentive (PLI) Scheme 2.0 for IT Hardware in India offers incentives, averaging around 5% on incremental sales, to eligible companies in the IT hardware manufacturing sector, India is well-equipped to seize opportunities and establish itself as a global leader in the IT hardware sector.



LED

Lighting segment in India is undergoing rapid transformation, with widespread adoption of LED lighting. LED lights are now extensively used in various domestic and industrial products, ranging from screens and walls to ceilings and wearables. LED lighting offers numerous advantages over conventional lighting systems, including energy efficiency, cost-saving, longer lifespan, and lower heat emission.

⁹ Ministry of Electronics and Information Technology, 2023, Annual Report

Sub-sectoral trends

Rise in the Indian Mobile Manufacturing

India has emerged as the second-largest global hub for mobile handset manufacturing and the world's second-largest smartphone market in FY22¹⁰. With production soaring from 6 crore units in 2014-15 to 29 crore units in 2020-21, the sector has become a major component of the "Make in India" initiative, hosting over 200 manufacturing units and employing approximately 7 lakh individuals.

China Leads Global Mobile Exports Market

China dominates the global mobile export market with an 67% share, followed by Vietnam at 13%, while India accounts for a 1% share. In terms of imports, the USA leads as the largest mobile importer at 19%, followed by Japan at 6% and Germany at 5%.

Top 5 key export economies for India

		Share in India's mobile exports	Strategic Shift: High-End Mobile Exports
UAE		23%	- India exports low end mobile phones (~Rs 9,000) while the top importing economies import mobile phones (>Rs 15,000). - Investment by foreign players under PLI scheme is expected to increase production and export of high-end mobile phones in India creating opportunities across top importing economies
USA		20%	
Netherlands		10%	
UK		8%	
		7%	

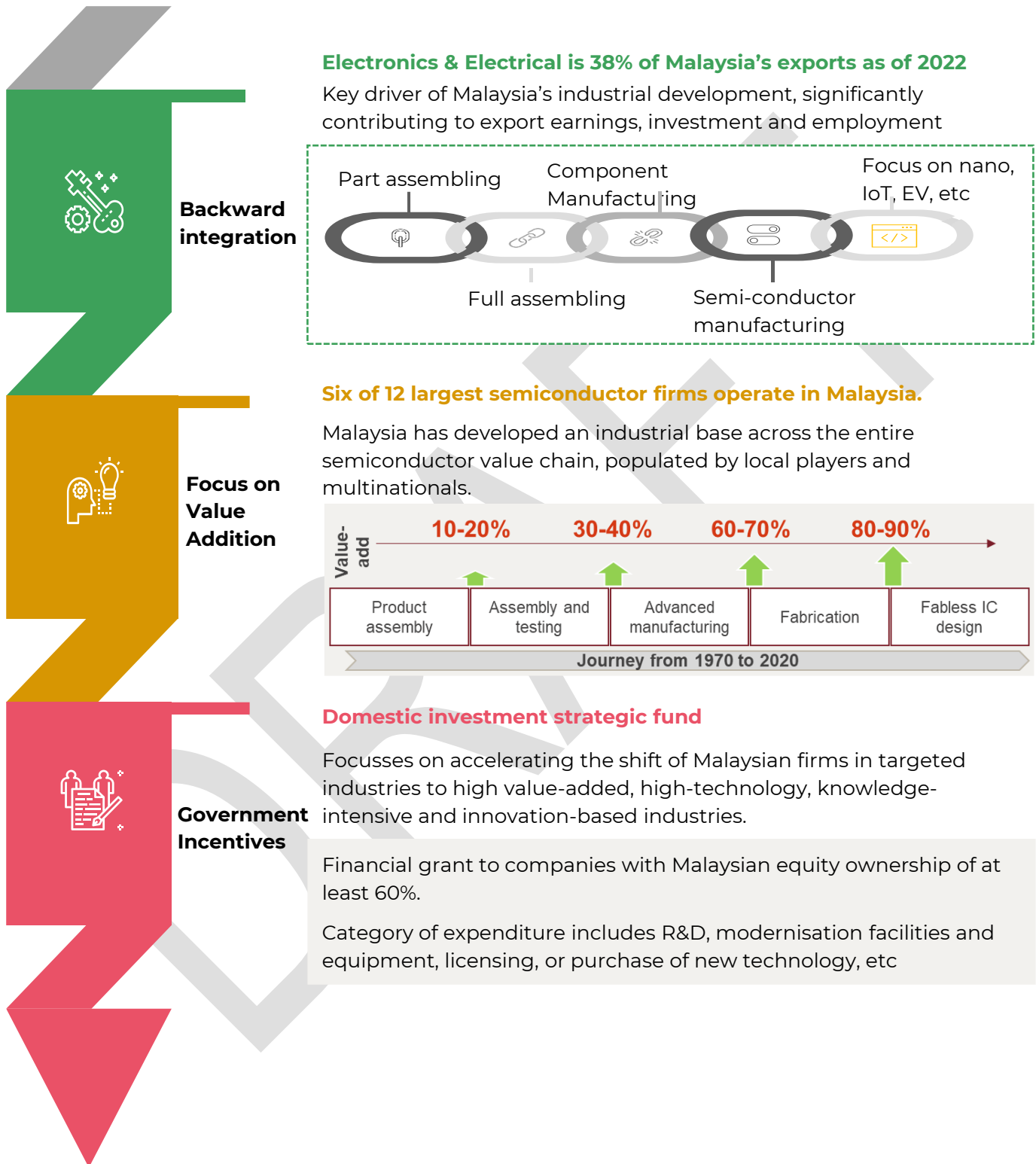
Note: Latest data as of FY2023. CBU stands for completely built units

Source: Trade data from Ministry of Commerce

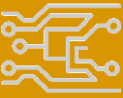
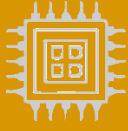
¹⁰ Ministry of Electronics and Information Technology, 2023, Annual Report

Malaysia's Electronic Manufacturing Boom: Case Study

Malaysia strategically transitioned from being a part assembler to a fabricator .by focusing on value addition across the chain as well as strategic investments by government in attracting global names and incentivizing domestic players investments in R&D/upgrade.



Sub-sectoral trends¹¹

For FY22E						
	Mobile Phones	Industrial electronics	Consumer electronics	Electronic components	IT/Computer Hardware	LED
Consumption	Rs. 2,505 billion	Rs 887 billion	Rs. 1,011 billion	Rs 2,470 billion	Rs 1,310 billion	Rs 237 billion
	CAGR 5-Years	CAGR 5-Years	CAGR 5-Years	CAGR 5-Years	CAGR 5-Years	CAGR 5-Years
	17%	5%	5%	15%	18%	15%
Production	Rs. 2,750 billion	Rs 820 billion	Rs 745 billion	Rs 708 billion	Rs 298 billion	Rs 186 billion
	CAGR 5-Years	CAGR 5-Years	CAGR 5-Years	CAGR 5-Years	CAGR 5-Years	CAGR 5-Years
	25%	6%	5%	6%	8%	21%
Import	Rs 122 billion	Rs 962 billion	Rs 301 billion	Rs 1,903 billion	Rs 1,048 billion	Rs 64 billion
	CAGR 5-Years	CAGR 5-Years	CAGR 5-Years	CAGR 5-Years	CAGR 5-Years	CAGR 5-Years
	-15%	10%	11%	19%	21%	5%
Export	Rs 357 billion	Rs 232 billion	Rs 35 billion	Rs 140 billion	Rs 36 billion	Rs 13 billion
	CAGR 5-Years	CAGR 5-Years	CAGR 5-Years	CAGR 5-Years	CAGR 5-Years	CAGR 5-Years
	99%	2%	2%	22%	7%	17%

Note: Sub-sectors considered constitute 80% of total electronics production as of FY22. Strategic electronics is excluded. Also, Imports not exhaustive, per MEITY data on total imports share.

Source:

Production: MEITY⁷

Consumption, Import & Export: Crisil Research, MEITY, Ministry of Commerce

¹¹ Ministry of Electronics and Information Technology, 2023, Annual Report

Challenges & Opportunities

Major Challenges



High dependence on imports

The Indian electronics industry faces significant challenge, in terms of high dependency on import for electronic components. The dependence increases the cost of production, as imported components are often subject to tariffs, taxes, and transportation costs, hence putting Indian manufacturers at a disadvantage, in terms of pricing their products competitively in the global market.

According to Department of Commerce, the import dynamics of electronic items, particularly Electronic Integrated Circuits and Micro-assemblies (HS Code: 8542) and items like LED and Piezo-electric crystals (HS Code: 8541), have undergone a notable shift. In FY18, these items constituted 5-7% of total electronics imports, but today, they account for 20-23%. China (including Hong Kong & Taiwan) emerges as the predominant importer, consistently holding a significant 70-75% share in the import of these items over the years.



Challenges in Building an Electronics Manufacturing Ecosystem

One of the significant challenges faced by the electronics industry in India is the lack of a well-established component ecosystem, which leads to heavy reliance on imports by manufacturers.

Additionally, the segment has several issues, including inadequate infrastructure, limitations with regard to the domestic supply chain and logistics, high finance costs, limited design capabilities, insufficient focus on R&D, and lack of skilled workforce.

Plug and play infrastructure⁸, as seen in competing nations like China and Vietnam, involves comprehensive support. Governments take charge of regulatory compliances, offering ready-made facilities, including on-site dormitories, to manufacturers.



Cost Disadvantages in India's Electronics Industry

India faces challenges in terms of higher import tariffs on input of electronic products compared with Asian peers. These tariffs are also subject to frequent amendments, creating a level of uncertainty for manufacturers.

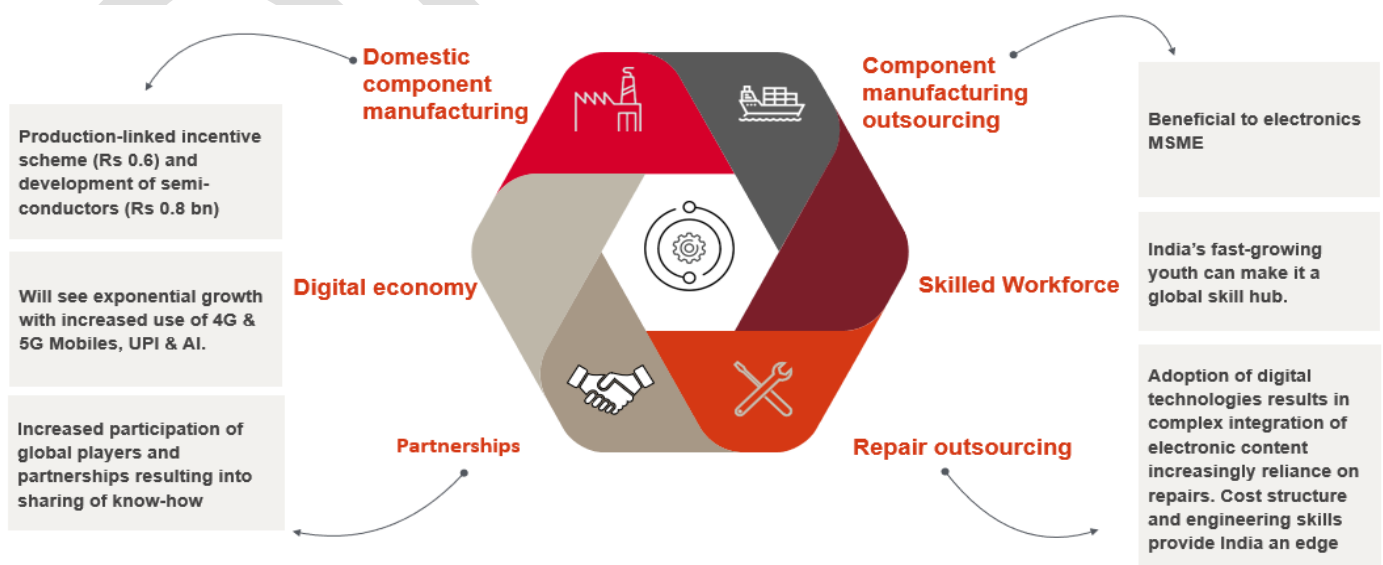
India contends with a cost disadvantage⁹ of 7-10% compared to Vietnam and 17-19% in comparison to China. This discrepancy arises from factors like land allocation, labor incentives, and research and development subsidies. This increased cost poses a challenge for Indian businesses, making them relatively more expensive than their counterparts and impacting their global competitiveness.

According to ICEA analysis¹⁰, the current customs tariff regime in India unfavourably impacts domestic electronics manufacturers compared to competing economies like China, Vietnam, Thailand, and Mexico. Higher tariffs could potentially result in inflated product costs, diminishing their competitiveness in the global markets.

Opportunities

The Indian government's introduction of PLI schemes, notably the one for Electronics Manufacturing covering mobile phones, IT hardware, and electronic components, offers manufacturers a strategic edge by incentivizing local component production. This reduces production costs and import dependency. India's mobile phone segment showcases this potential, making it the world's second-largest mobile handset manufacturer and the second-largest smartphone market, solidifying its status as the fastest-growing globally, promising immense possibilities for electronics manufacturers. India's electronics industry can expand globally by emphasizing quality production and competitive pricing, exploring export opportunities. To capitalise on domestic growth, manufacturers can tailor products to Indian preferences and price points, tapping into the expanding consumer base driven by rising disposable incomes and the affluent segment.

As the Indian electronics industry grows, there's an emerging opportunity to explore repair outsourcing as another viable field



Government Interventions

The government of India has rolled out a comprehensive array of initiatives and schemes to empower the electronics manufacturing sector. These encompass various facets, including production incentives, export promotion, research and development support, and ease of doing business measures. These initiatives are geared towards bolstering the electronics manufacturing industry's competitiveness, fostering growth, and expanding its global footprint.

Major Policies & Schemes¹²

National Policy on Electronics (NEP), 2019

The NPE 2019 policy provides a roadmap for the long-term development of the electronics sector in India. By focusing on domestic manufacturing, research, and development (R&D), export promotion, skill development, and manufacturing of components, the policy aims to transform India into a globally competitive electronics manufacturing hub and drive the sector's contribution to the economy.

Modified Electronics Manufacturing Clusters (EMC 2.0) Scheme, 2020

The scheme focuses on attracting major global electronic manufacturers, along with their supply chains, to set up production units in India & development of World class Plug and Play infrastructure along with Standard Factory Sheds. By attracting large-scale electronic manufacturers as anchor units, the scheme seeks to bring in their suppliers and create robust manufacturing clusters.

The total outlay of the propose EMC 2.0 Scheme is Rs. 3,762 crores. Under the scheme, three applications for setting up electronic manufacturing clusters with total project cost of Rs 1,903 crore, including central financial assistance of Rs 889 crores, have been approved till September 2022.

Production Linked Incentive (PLI) scheme: Large Scale Electronics Manufacturing, 2020

The scheme provides financial incentives to eligible companies to encourage the production of manufactured goods, including mobile phones and specified electronic components.

Under the scheme, eligible companies receive incentives ranging from 4-6% on incremental sale of manufactured goods over the fiscal 2020 base year for five years. The total outlay under the Scheme is ₹ 38,645 crore. The scheme focuses on promoting large-scale electronics manufacturing, particularly in the mobile phones segment. Total expenditure outlay for this scheme is USD 5.7 billion till fiscal 2025.

Major Policies & Schemes¹³

Scheme for Promotion of Manufacturing of Electronic Components and Semiconductors (SPECS), 2020

Aimed at promoting the development of the electronic components manufacturing ecosystem in India. The scheme provides a financial incentive of 25% on capital expenditure for a specified list of electronic goods that form the downstream value chain of electronic products.

This includes electronic components, semiconductor/display fabrication units, ATMP (Assembly, Testing, Marking, and Packaging) units, specialized sub-assemblies, and capital goods required for manufacturing these goods.

PLI scheme: IT Hardware, 2021

The scheme aims to stimulate domestic manufacturing and attract significant investment in the IT

¹² Ministry of Electronics and Information Technology, 2023, Production Linked Incentive Scheme

¹³ Ministry of Electronics and Information Technology, 2023, Annual Report

hardware sector.

It provides incentives ranging from 1% / 2% to 4% on net incremental sales of goods in the target segments, such as laptops, tablets, all-in-one personal computers, and servers. The total outlay under the Scheme is ₹ 7,350 crore. The scheme is expected to bring additional investment in electronics manufacturing of Rs 2,500 crore.

Empowering India's Semiconductor Industry¹⁴

The India Semiconductor Mission (ISM) is a vital autonomous initiative under the Digital India Corporation, focused on invigorating India's semiconductor ecosystem, spanning manufacturing, packaging, and design. With financial and administrative autonomy and guidance from a global advisory board, ISM aims to formulate a sustainable semiconductor strategy, foster secure microelectronics adoption, boost the Indian semiconductor design sector, promote indigenous IP creation, enable tech transfers, and drive cutting-edge research. Under the Semicon India Programme, ISM oversees four key schemes, striving to nurture India's semiconductor and display manufacturing ecosystem with a total budget of INR 76,000 crore.

The government's paramount objective is strengthening the semiconductor ecosystem to propel India's burgeoning electronics manufacturing and innovation landscape. With an initial INR 13 crore allocation, this initiative incentivizes investments in semiconductors, display manufacturing, and design. Additionally, modernizing the Semi-Conductor Laboratory in Mohali further cements India's position in the global semiconductor industry, positioning the nation as a pivotal player in the global electronics value chain.

¹⁴ Press Information Bureau, 2022

Schemes under The India Semiconductor Mission



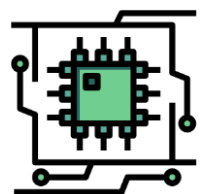
Scheme for Semiconductor Fabs, 2022

Encourages significant investments in semiconductor wafer fabrication facilities, offering 50% fiscal support for Silicon CMOS-based Semiconductor Fabs' establishment in India, strengthening the electronics manufacturing ecosystem.



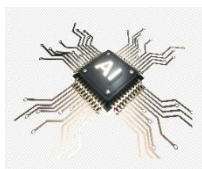
Scheme for Display Fabs, 2022

Promotes TFT LCD or AMOLED based display panel manufacturing in India, providing 50% fiscal support for project costs, fostering the growth of Display Fabs within the country.



Scheme for Compound Semiconductors and More, 2021

Extends fiscal support to various semiconductor endeavors, covering 50% of the Capital Expenditure for Compound Semiconductors, Silicon Photonics (SiPh), Sensors (including MEMS), Discrete Semiconductors Fab, and Semiconductor Assembly, Testing, Marking, and Packaging (ATMP) or OSAT facilities, facilitating diverse semiconductor production



Semicon India Future Design (DLI) Scheme, 2021

Supports semiconductor design activities with financial incentives and infrastructure support across the design and deployment phases, including "Product Design Linked Incentive" and "Deployment Linked Incentive" components, benefiting Integrated Circuits (ICs), Chipsets, System on Chips (SoCs), Systems & IP Cores, and semiconductor-linked designs.

Way forward



Strengthen the component ecosystem

Focus on developing a robust and self-reliant electronic component manufacturing ecosystem in India to reduce dependence on imports and enhance competitiveness.

As per the Trade data from Ministry of Commerce a, around 49% of the total electronics component imports are from China



Enhance research and development capabilities

Investment in R&D to foster innovation, develop cutting-edge technologies and drive product development in the electronics industry.

According to a Meity report¹⁵, R&D subsidies in India are notably lower, at 0.15%, compared to the range of 0.4-1% in Vietnam and 2% in China, impacting the country's cost-reduction efforts.



Improve ease of doing business

Streamline regulations, reduce bureaucratic hurdles, and enhance infrastructure to improve the ease of doing business for electronics manufacturing, fostering industry growth and investment.

According to a ICEA report¹⁶, improving the ease of doing business is a key factor in the growth of the electronics market in India. Factors influencing the "Ease of doing business" in India create a cost disadvantage for electronics manufacturing compared to China by 2-3% and Vietnam by 1.5-2.5%.



Export competitiveness & align with global standards and certifications

Developing strategies to boost exports of electronic products by improving quality standards, exploring new markets, and leveraging international trade agreements.

Ensure compliance with international standards and certifications to enhance the credibility and acceptance of Indian electronic products in global markets.

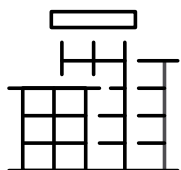
According to a Niti aayog report¹⁷, maintaining high quality and adhering to

¹⁵ Ministry of Electronics and Information Technology, R&D in Electronics

¹⁶ ICEA. (2022). Aatmanirbhar Bharat: India's Turning Point

¹⁷ NITI Aayog, 2022, Export Preparedness Index 2022

	international standards and certifications is crucial for gaining the trust of global buyers
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Establishing Electronics-Specific Special Economic Zones (SEZs)

Developing dedicated SEZs tailored to the needs of the electronic manufacturing sector can create an environment conducive to rapid industry growth and relocation. These SEZs can provide various incentives, such as tax benefits, streamlined regulations, and infrastructure support, to attract foreign direct investment and promote local manufacturing.

According to a ICEA report, SEZs allow for a faster set up of manufacturing plants and ease of doing business

Recommended Readings

1. Ministry of Electronics and Information Technology. (2023). Annual Report 2022-23
https://www.meity.gov.in/writereaddata/files/AR_2022-23_English_24-04-23.pdf
2. Press Information Bureau. (2022, January). Vision \$300 billion. Government of India.
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